Category Theory

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Big Picture / Motivation

- Category theory is a "theory", like set theory, like group theory, like type theory
- Types in functional programming $\rightsquigarrow$ sets (or set-like objects)
- Functions $\rightsquigarrow$ arrows/morphisms between types
- Category theory gives language to talk about how we compose computations, abstract patterns, and universal constructions that mirror programming constructs (pairs, function types, currying)
- It provides means of abstraction to reason and prove things generically, instead of on case-by-case basis

Course Outline

Part I: Sets and Warm-Up

Sets, functions, isomorphisms, quotients and equivalences

Part II: Categories & Functors

Definitions and examples, products, coproducts, commutative diagrams, universal properties

Part III: Natural Transformations

Natural transformations, exponentials, Yoneda Theorem

Part IV: Adjunctions & Free Objects

Adjunctions, free objects, initial algebras, induction

These slides:

www.sergey-goncharov.org/mgs-2026

Recommended Sources

- **Steve Awodey** — *Category Theory*, 2nd ed., Oxford University Press, 2010
  - [accessible introduction]
- **Emily Riehl** — *Category Theory in Context*, Dover, 2016
  - [freely available at https://math.jhu.edu/~eriehl/161/context.pdf]
- **Roy Crole** — *Categories for Types*, Cambridge University Press, 1993
  - [CS-oriented; Category theory for types & denotational semantics]
- **Saunders Mac Lane** — *Categories for the Working Mathematician*, 2nd ed., Springer, 1998
  - [timeless classics]

Sets and Warm-Up

Why Begin with Sets?

- Mathematics builds on sets
- Types in programming follow set intuition
- Category theory (standardly) also builds on sets
- But: what exactly is a set?
  - OK, that is too ambitious – what is it roughly?
- We start with sets to discover something deeper is hiding underneath

Naïve (Philosophical) Sets

- Set is collection of objects, like

$$X = \left\{ \text{⚽},\ \text{🐕},\ \pi,\ \mathbb{N},\ \text{♥} \right\}$$

- We write $x \in X$ to say that x is member of X, e.g. $\pi \in X$
- Empty set $\varnothing = \{\ \}$ is a set of no elements
- There are various mechanisms to construct larger sets
- Set comprehension

$$\{x \in S \mid P(x)\} \qquad \text{e.g.} \qquad \text{evens} = \{n \in \mathbb{N} \mid \exists m \in \mathbb{N}.\, n = 2m\}$$

Intuition: Sets are containers

Finite Sets v.s. Finite Lists

- (Finite) lists over set X: $\mathrm{List}(X) = \{[x_1, \ldots, x_n] \mid n \in \mathbb{N}, x_1 \in X, \ldots, x_n \in X\}$
- **Notation:** $[\,]$ – empty list, $x :: xs$ – list with head x and tail xs. E.g. $x :: [\,] = [x]$
- Finite sets are lists, where we additionally enable
  - Commutativity: $[\ldots, x, y, \ldots] \equiv [\ldots, y, x, \ldots]$
  - Idempotence: $[\ldots, x, x, \ldots] \equiv [\ldots, x, \ldots]$
- (Finite) Multisets = lists with commutativity, but not idempotence. Thus

$$\text{Lists} \quad < \quad \text{Multisets} \quad < \quad \text{Sets}$$

Sets (Slightly More) Formally

- Set theory postulates: everything is a set
- Only primitive relation between sets is '$\in$'

We then derive:

- Natural numbers: $0 = \varnothing$, $1 = \{0, \varnothing\} = \{\varnothing\}$, $2 = \{1, \varnothing\} = \{\{\varnothing\}, \varnothing\}, \ldots$
- Set inclusion: $X \subseteq Y$ if $\forall x \in X. x \in Y$
  - Extensionality axiom: $X = Y$ if and only if $X \subseteq Y$ and $Y \subseteq X$
- Powerset: $Y \in \mathcal{P}(X)$ if and only if $Y \subseteq X$
- Relations: $R \subseteq X \times Y$
- Functions: $f : X \to Y$ – such relations $R \subseteq X \times Y$ that for every $x \in X$ there is unique $y \in Y$ with $x \, R \, y$
  - So, in set theory: Sets $\rightsquigarrow$ Relations $\rightsquigarrow$ Functions

Empty Set and Singleton Sets

- No controversy about empty set: $\varnothing = \{\ \}$ – set with no elements
- But what is one-element set? {⚽}? {🐕}? $\{\pi\}$?
- $\{\varnothing\}$ is good candidate
- For every set S, $\{S\}$ is also singleton set
- Singleton sets are characterized by property: they are sets with exactly one element
- $\{\varnothing\}$ is a concrete representative in the class of all singletons

Note on Classes

- All sets do not form a set, but a class

$$\mathsf{Set} < \mathsf{Class}$$

- Division into sets and proper classes is way to avoid paradoxes
- Other approach: universes

$$\mathsf{Set}_0 < \mathsf{Set}_1 < \mathsf{Set}_2 < \dots$$

  where the union of all sets in Set_i is in Set_{i+1}
- By saying "set" we can imagine set from Set_i, and by saying "class" we can imagine set from Set_{i+1}

Natural Numbers as Sets

- In computer science, we count from 0: $\mathbb{N} = \{0, 1, \ldots\}$!
- Von Neumann numerals: $0 := \varnothing, n+1 := n \cup \{n\}$ So:

$$0 = \varnothing, \quad 1 = \{\varnothing\}, \quad 2 = \{\varnothing, \{\varnothing\}\}, \quad 3 = \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}, \ldots$$

  This is very convenient, because $n+1 = \{0, \ldots, n\}$
- Alternatively Zermelo numerals: $0 := \varnothing, n+1 := \{n\}$. This is less convenient

Ordered Pairs and Products

We would like to define Cartesian products $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$. But what is (x, y)?

- Kuratowski definition: $(x, y) := \big\{\{x\}, \{x, y\}\big\}$
- Then (PROVE IT) :

$$(x, y) = (x', y') \iff x = x' \quad \text{and} \quad y = y'.$$

- Using this, we can define:

$$X \times Y := \Big\{\{\{x\}, \{x, y\}\} \subseteq \mathcal{P}(X \cup Y) \mid x \in X,\ y \in Y\Big\}$$

Pairs and Cartesian products are constructed objects!

Disjoint Unions

- Union of sets: $x \in X \cup Y$ iff $x \in X$ or $x \in Y$
- How to define Disjoint Union $X + Y$ (= $X \uplus Y$)?
- Maybe $X + Y = \{(x, \text{🎸}) \mid x \in X\} \cup \{(x, \text{🔨}) \mid y \in Y\}$?

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- Maybe $X + Y = \{(x, \text{🍔}) \mid x \in X\} \cup \{(x, \text{🐎}) \mid y \in Y\}$?
- More seriously: $X + Y = \{(x, 0) \mid x \in X\} \cup \{(x, 1) \mid y \in Y\}$

Now compute with von Neumann numerals: $1 = \{0\}$, $2 = \{0, 1\}$. Then

$$1 + 1 = \{(1, 0), (1, 1)\} = \{\{\{1\}, \{1, 0\}\}, \{\{1\}\}\}$$

This is not equal to $2 = \{0, 1\}$! So

$$1 + 1 \neq 2$$

Functions between Sets

- Function $f : X \to Y$, assigns each $x \in X$ a unique $f(x) \in Y$
- Functions $f : X \to Y$ are special relations $f \subseteq X \times Y$
- Identity function $id_X : X \to X$, $id_X(x) = x$
- Equality of functions $f = g$ if for all x, $f(x) = g(x)$
  - This is provable principle (PROVE IT) , called function extensionality

Composition:

$$(g \circ f)(x) = g(f(x))$$

Proposition: $f \circ id = f$, $id \circ f = f$ and $f \circ (g \circ h) = (f \circ g) \circ h$ (associativity)

Bijection vs Isomorphism

Function $f : X \to Y$ is

- Injection if $f(x) = f(x')$ entails $x = x'$
- Surjection if for every $y \in Y$ there is $x \in X$, such that $f(x) = y$
- Bijection if it is both injection and surjection

Alternative view: $f : X \to Y$ is isomorphism if for some $f^{-1} : Y \to X$,

$$f \circ f^{-1} = id_Y \quad \text{and} \quad f^{-1} \circ f = id_X$$

Proposition: f is bijection if and only if f is isomorphism

Some Properties of Isomorphisms

It follows that (PROVE IT)

- f^{-1} is unique if exists
- If f is isomorphism, so is f^{-1}
- id is isomorphism and $id^{-1} = id$
- Composition of isomorphisms $f \circ g$ is isomorphism and $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

These are derivable from definition, without using that isomorphism = bijection!

Notation: $X \cong Y$ if there is isomorphism $f : X \to Y$

Up-to Isomorphism

- Big innovation of category theory: work with things via their descriptions
- We narrow down objects of interest as those that "fit best" to such description
- Such descriptions together with their best-fit condition are called universal properties
- They determine objects not uniquely, but up-to isomorphism

Example. A singleton set 1 is characterised by: there is exactly one function $X \to 1$ for every set X

- $\{\varnothing\}$, $\{\{\varnothing\}\}$, $\{$⚽$\}$ all satisfy this — and are all isomorphic
- Any two singletons are isomorphic, so we speak of the singleton up to isomorphism

Induction/Recursion for Natural Numbers

- Set theory postulates existence of natural numbers $\mathbb{N}$ as smallest set that contains $0 = \varnothing$ and closed under $n \mapsto n + 1 := n \cup \{n\}$
- Induction is paraphrasing this: if $P \subseteq \mathbb{N}$ with $0 \in P$ and $n \in P \Rightarrow n{+}1 \in P$, then $P = \mathbb{N}$
  - Because predicates over $\mathbb{N} \iff$ subsets of $\mathbb{N}$:

$$P(0) \wedge (\forall n.\, P(n) \Rightarrow P(n+1)) \Rightarrow \forall n.\, P(n)$$

- (Structural) Recursion: for any $a \in A$ and $h\colon A \to A$, there is unique $f : \mathbb{N} \to A$ with

$$f(0) = a \qquad\qquad f(n{+}1) = h(f(n))$$

- Primitive recursion is derivable from recursion (PROVE IT) : for any $g\colon X \to A$ and $h\colon A \times X \times \mathbb{N} \to A$, there is unique $f\colon X \times \mathbb{N} \to A$ with

$$f(x, 0) = g(x) \qquad\qquad f(x, n{+}1) = h(f(x, n), x, n)$$

Induction/Recursion for Lists

- Given set X, $\mathrm{List}(X)$ is constructable as smallest set that contains $[\,]$ and closed under $xs \mapsto x :: xs$ for $x \in X$
- Induction: $P([\,]) \wedge (\forall x \in X, xs.\, P(xs) \Rightarrow P(x :: xs)) \Rightarrow \forall xs.\, P(xs)$
- (Structural) Recursion: for any $a \in A$ and $h\colon X \times A \to A$, there is unique $f\colon\ \mathrm{List}(X) \to A$ with

$$f([\,]) = a \qquad\qquad f(x :: xs) = h(x, f(xs))$$

Questions:

- What is primitive recursion for lists?
- Why it is derivable (PROVE IT)
- Define *length* and concatenation recursively

Remarks:

- Structural recursion $\neq$ general recursion (no arbitrary self-calls)
- Induction — proof principle; recursion — definition principle

Categories & Functors

Definition of Category

Category $\mathcal{C}$ consists of:

- Class of objects $\mathrm{Ob}(\mathcal{C})$
- For every A, B set of morphisms $\mathrm{Hom}_{\mathcal{C}}(A, B)$. Write $f : A \to B$
- Composition: if $f : A \to B$ and $g : B \to C$ then $g \circ f : A \to C$
- Identity morphisms $id_A : A \to A$

Subject to associativity $(h \circ g) \circ f = h \circ (g \circ f)$ and identity laws $id \circ f = f = f \circ id$

We can imagine categories as (multi)graphs: objects = nodes, morphisms = edges

Sets as Category

Category of sets **Set**

- **Objects:** sets
- **Morphisms:** functions between sets
- Composition: usual function composition
- Identities: identity functions

Further set-like categories (objects = sets, identities = identity functions):

- Relations (**Rel**) – morphisms are relations
- Partial functions
- Injections
- Isomorphisms

...

Category Pos

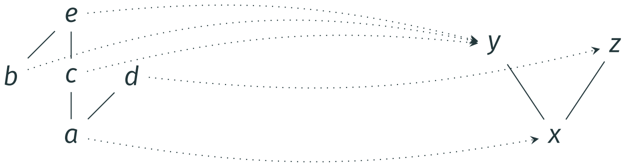
- **Objects:** partially ordered sets $(P, \leq)$
- **Morphisms:** monotone functions

$$f : (P, \leq_P) \to (Q, \leq_Q)$$

such that

$$x \leq_P y \Rightarrow f(x) \leq_Q f(y)$$

Example:



Category Mon

- **Objects:** monoids $(M, \cdot, e)$
- **Morphisms:** monoid homomorphisms are maps $f\colon M \to N$, such that unit and multiplication are preserved:

$$f(x \cdot y) = f(x) \cdot f(y) \qquad\qquad f(e_M) = e_N$$

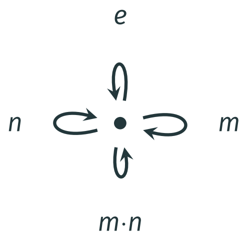
Example: $\mathit{length}\colon (\mathrm{List}(A), +\!\!+, [\,]) \to (\mathbb{N}, +, 0)$ — preserves unit and multiplication:

$$\begin{aligned} \mathit{length}([\,]) &= 0 \\ \mathit{length}(xs +\!\!+ ys) &= \mathit{length}(xs) + \mathit{length}(ys) \end{aligned}$$

Monoid as One-Object Category

Fix some monoid $(M, \cdot, e)$. We can turn it into category!

- **Objects:** — exactly one, say $\bullet$
- **Morphisms:** $\mathrm{Hom}(\bullet, \bullet) = M$
- Composition is given by monoid multiplication
- Identity given by e



This is equivalence: Monoids $\iff$ one-object categories (PROVE IT)

Monoid homomorphisms $\iff$ functors between one-object categories (PROVE IT)

Posets as Categories

Fix poset $(P, \leq)$. It is also category:

- **Objects:** elements of P
- **Morphisms:** $\mathrm{Hom}(x, y) = \{\star\}$ if $x \leq y$ and $\mathrm{Hom}(x, y) = \varnothing$ otherwise (at most one morphism between two objects)
- Unit of composition = reflexivity
- Associativity of composition = transitivity

Hint: Categorical notions become order-theoretic when instantiated to poset-categories.

To understand a complex categorical notion, instantiate it to poset-category!

Example Categories: Summary

Category	Objects	Morphisms	Composition
Set	sets	functions	function composition
Pos	posets	monotone maps	function composition
Mon	monoids	homomorphisms	function composition
Rel	sets	relations $R \subseteq X \times Y$	relational composition
$(P, \leq)$	elements of P	$\star$ if $x \leq y$	transitivity
$(M, \cdot, e)$	$\{\star\}$	elements of M	monoid multiplication

Commutative Diagrams

- For $f : A \to B$, we can express $f \circ id_A = f$ and $id_B \circ f = f$ diagrammatically:

$$\begin{tikzcd}
 & A \arrow[r, "f"] \arrow[d, "f"] & B \\
A \arrow[ur, equal] \arrow[r, "f"] & B \arrow[ur, equal] &
\end{tikzcd}$$

- Associativity:

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] & B \arrow[r, "g"] \arrow[rr, bend right, "h \circ g"'] & C \arrow[r, "h"] & D
\end{tikzcd}$$

But that is redundant, because we read sequence f, g as $g \circ f$ – associativity is baked in!

Isomorphisms

Morphism $f : A \to B$ is an isomorphism if there exists $f^{-1} : B \to A$ with $f^{-1} \circ f = id_A$ and $f \circ f^{-1} = id_B$:

$$\begin{tikzcd} A \arrow[loop left, "id_A"] \arrow[r, bend left, "f"] & B \arrow[l, bend left, "f^{-1}"] \arrow[loop right, "id_B"] \end{tikzcd}$$

- A and B are isomorphic, written $A \cong B$, if such f exists
- f^{-1} is unique: if $g \circ f = id_A$ and $f \circ g = id_B$, then

$$g = g \circ id_B = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = id_A \circ f^{-1} = f^{-1}$$

- In **Set**: isomorphism = bijection; in **Pos**: order-isomorphism; in **Mon**: monoid isomorphism
- Group is one-object category in which every morphism is isomorphism (PROVE IT)

Commutative Diagrams and Equational Reasoning

A diagram commutes if every directed path with the same start and end gives the same composite.

$$
\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "a"'] & B \arrow[r, "g"] \arrow[d, "b"] & C \arrow[d, "c"] \\
D \arrow[r, "h"'] & E \arrow[r, "k"'] \arrow[d, "p"'] & F \arrow[d, "q"] \\
& G \arrow[r, "r"'] & H
\end{tikzcd}
$$

If all three squares commute:

$$b \circ f = h \circ a, \quad c \circ g = k \circ b, \quad q \circ k = r \circ p$$

then the outer paths from A to H agree:

$$
\begin{aligned}
q \circ c \circ g \circ f &= q \circ (k \circ b) \circ f \\
&= (q \circ k) \circ (b \circ f) \\
&= (r \circ p) \circ (h \circ a) \\
&= r \circ p \circ h \circ a
\end{aligned}
$$

Thus, commutative diagrams = pictorial representations of equational proofs!

Not Every Diagram Proves Something

$$
\begin{tikzcd}
A \arrow[rr, "f"] \arrow[dd, "g"'] & & B \arrow[d, "h"] \\
& C \arrow[ul, "u"'] \arrow[r, "k"] \arrow[d, "j"'] & D \arrow[d, "r"] \\
E \arrow[r, "p"] & F \arrow[r, "q"] & G
\end{tikzcd}
$$

Commuting cells (by assumption):

$$h \circ f \circ u = k$$

$$p \circ g \circ u = j$$

$$r \circ k = q \circ j$$

Chaining all three:

$$
\begin{aligned}
(r \circ h \circ f) \circ u &= r \circ k \\
&= q \circ j \\
&= (q \circ p \circ g) \circ u
\end{aligned}
$$

But this does not imply $r \circ h \circ f = q \circ p \circ g$, because we generally cannot discard u!

Terminal and Initial Objects

In category $\mathcal{C}$:

- Terminal object 1: for every object A there is exactly one morphism $A \to 1$
- Initial object 0: for every object A there is exactly one morphism $0 \to A$

Examples:

- In **Set**: any singleton $\{*\}$ is terminal; $\varnothing$ is initial
- In **Pos**: any one-element poset is terminal; $\varnothing$ is initial
- In **Mon**: trivial monoid $\{e\}$ is terminal and initial
- In every poset category: terminal = greatest element; initial = least element

Key viewpoint: elements of X in **Set** correspond to arrows $1 \to X$, i.e. elements$(X) \cong \mathrm{Hom}(1, X)$. So, $f(x)$ must be interpreted as composition $1 \xrightarrow{x} X \xrightarrow{f} Y$!

Product Universal Property

- **Object:** a product of X and Y is an object Z
- **Diagram (span):** morphisms $p_1 \colon Z \to X$ and $p_2 \colon Z \to Y$
- **Universal property:** for any $f : A \to X$ and $g \colon A \to Y$ there is a unique $h \colon A \to Z$ with $p_1 \circ h = f$ and $p_2 \circ h = g$:

$$\begin{tikzcd} & A \arrow[dl, "f"'] \arrow[d, dashed, "h"] \arrow[dr, "g"] & \\ X & Z \arrow[l, "p_1"] \arrow[r, "p_2"'] & Y \end{tikzcd}$$

Selected product $X \times Y$: specific choice of product for every pair (X, Y); all products of X and Y are uniquely isomorphic

Notation: With $X \times Y$, use $\mathit{fst}, \mathit{snd}$ for the chosen projections; $\langle f, g \rangle$ for the induced h

Examples of Products

- **Set:** $X \times Y = \{(x, y) \mid x \in X,\ y \in Y\}$; projections $(x, y) \mapsto x$ and $(x, y) \mapsto y$; pairing $\langle f, g\rangle(x) = (f(x), g(x))$
- **Pos:** $X \times Y$ with pointwise order: $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$
- **Mon:** componentwise monoid $(M \times N,\ \cdot,\ (e_M, e_N))$ where

$$(m, n) \cdot (m', n') = (m \cdot m',\ n \cdot n')$$

- **Poset category:** product of x and y = meet $x \wedge y$ (=greatest lower bound), when it exists

Products are Up-To-Isomorphism

Let (Z, p_1, p_2) and (Z', p_1', p_2') both be products of X and Y:

- Universal property of Z: $\exists!\, i : Z' \to Z$ with $p_1 \circ i = p_1'$, $p_2 \circ i = p_2'$
- Universal property of Z' on span (p_1, p_2): $\exists!\, j : Z \to Z'$ with $p_1' \circ j = p_1$, $p_2' \circ j = p_2$
- By uniqueness: $j \circ i = \mathit{id}_{Z'}$
- By symmetric argument: $i \circ j = \mathit{id}_Z$, so $Z \cong Z'$

$$\begin{tikzcd}
 & Z' \arrow[dl, "p_1'"'] \arrow[d, dashed, "i"] \arrow[dr, "p_2'"] & \\
X & Z \arrow[l, "p_1"'] \arrow[r, "p_2"] \arrow[d, dashed, "j"] & Y \\
 & Z' \arrow[ul, "p_1'"] \arrow[ur, "p_2'"'] &
\end{tikzcd}$$

Example: Take selected product $(X \times Y, \mathit{fst}, \mathit{snd})$. Then $(Y \times X, \mathit{snd}, \mathit{fst})$ is also product of X and Y – isomorphism swap $= \langle \mathit{snd}, \mathit{fst} \rangle : X \times Y \xrightarrow{\sim} Y \times X$

Axioms of Binary Products

For selected products, one can show (PROVE IT) :

$$(\beta_1) \quad \mathit{fst} \circ \langle f, g \rangle = f$$

$$(\beta_2) \quad \mathit{snd} \circ \langle f, g \rangle = g$$

$$(\eta) \quad \langle \mathit{fst}, \mathit{snd} \rangle = \mathit{id}_{X \times Y}$$

$$(\mathrm{nat}) \quad \langle f, g \rangle \circ h = \langle f \circ h,\ g \circ h \rangle$$

Theorem: This is a sound and complete axiomatization of binary products (PROVE IT)

Example (Uniqueness of pairing): if $\mathit{fst} \circ k = f$ and $\mathit{snd} \circ k = g$, then

$$k \;=\; \mathit{id} \circ k \;\overset{(\eta)}{=}\; \langle \mathit{fst}, \mathit{snd} \rangle \circ k \;\overset{(\mathrm{nat})}{=}\; \langle \mathit{fst} \circ k,\ \mathit{snd} \circ k \rangle \;\overset{(\mathrm{assm.})}{=}\; \langle f, g \rangle$$

Dual Category

Given category $\mathcal{C}$, the opposite category $\mathcal{C}^{\text{op}}$ has:

- Same objects as $\mathcal{C}$
- Morphisms reversed: $f : A \to B$ in $\mathcal{C}$ becomes $f^{\text{op}} : B \to A$ in $\mathcal{C}^{\text{op}}$
- Composition: $g^{\text{op}} \circ f^{\text{op}} = (f \circ g)^{\text{op}}$

Principle of Duality: any theorem proved in $\mathcal{C}$ yields a dual theorem in $\mathcal{C}^{\text{op}}$ — just reverse all arrows

Examples:

- Poset $(P, \leq)^{\text{op}}$ = poset with reversed order $(P, \geq)$
- Initial objects in $\mathcal{C}$ = terminal objects in $\mathcal{C}^{\text{op}}$

Coproduct via Duality

Coproduct = product in $\mathcal{C}^{\mathrm{op}}$ — obtained by reversing all arrows in the product definition

- **Object:** a coproduct of X and Y is an object Z
- **Diagram (cospan):** morphisms $\iota_1 : X \to Z$ and $\iota_2 : Y \to Z$
- **Universal property:** for any $f : X \to A$ and $g : Y \to A$ there is a unique $h : Z \to A$ with $h \circ \iota_1 = f$ and $h \circ \iota_2 = g$:

$$\begin{tikzcd}
& A & \\
X \arrow[ur, "f"] \arrow[r, "\iota_1"'] & Z \arrow[u, dashed, "h"] & Y \arrow[ul, "g"'] \arrow[l, "\iota_2"]
\end{tikzcd}$$

Selected coproduct $X + Y$: choice of coproduct for every pair; notation $\mathit{inl}, \mathit{inr}$ for chosen injections; $[f, g]$ — co-pairing — for the induced h

Example: In **Set**, $X + Y$ = disjoint union (PROVE IT)

Axioms of Binary Coproducts

For selected coproducts, one can show:

$$(\beta_1)\quad [f, g] \circ \mathit{inl} = f$$

$$(\beta_2)\quad [f, g] \circ \mathit{inr} = g$$

$$(\eta)\quad [\mathit{inl}, \mathit{inr}] = \mathit{id}_{X+Y}$$

$$(\mathrm{nat})\quad h \circ [f, g] = [h \circ f,\ h \circ g]$$

Theorem: This is sound and complete axiomatization of binary coproducts

Proof: Duality

Example: Every $f : X + Y \to Z$ has form $[f_1, f_2]$ for $f_1 : X \to Z$, $f_2 : Y \to Z$. Indeed:

$$f \;=\; f \circ \mathit{id} \overset{(\eta)}{=} f \circ [\mathit{inl}, \mathit{inr}] \overset{(\mathrm{nat})}{=} [f \circ \mathit{inl},\ f \circ \mathit{inr}]$$

Categories & Functors

Functors

Definition of Functor

(Covariant) Functor $F \colon \mathcal{C} \to \mathcal{D}$ acts on objects and morphisms as follows:

$$A \in \mathrm{Ob}(\mathcal{C}) \mapsto F(A) \in \mathrm{Ob}(\mathcal{D}) \qquad \begin{tikzcd} A \arrow[d, "f"'] & & F(A) \arrow[d, "F(f)"] \\ B & \longmapsto & F(B) \end{tikzcd}$$

preserving composition and identities:

$$F(id_A) = id_{F(A)} \qquad F(g \circ f) = F(g) \circ F(f)$$

- Functor $F \colon \mathcal{C} \to \mathcal{C}$ is called endofunctor
- Functors compose: $(GF)(A) = G(F(A))$, $(GF)(f) = G(F(f))$; categories and functors form **Cat** — category of categories, modulo size issues

Examples of Functors

Endofunctors on Set:

- Powerset: $\mathcal{P}(X)$; on $f : X \to Y$ acts as direct image $\mathcal{P}(f)(S) = \{f(x) \mid x \in S\}$
- List: List(X) = finite lists over X; List(f)$([x_1, \ldots, x_n]) = [f(x_1), \ldots, f(x_n)]$
- Maybe: $X \mapsto X + \{\text{Nothing}\}$:
  - Maybe$(f)(\mathit{inl}\, x) = \mathit{inl}(f(x))$
  - Maybe$(f)(\mathit{inr}\, x) = \mathit{inr}\, x$
- Product: $X \mapsto A \times X$ for fixed A

Between different categories:

- Forgetful functor $U : \mathbf{Mon} \to \mathbf{Set}$: sends $(M, \cdot, e)$ to M, homomorphisms to functions
- Free monoid functor $F : \mathbf{Set} \to \mathbf{Mon}$: sends X to List(X)

Bifunctors

Bifunctor $F\colon \mathcal{C} \times \mathcal{D} \to \mathcal{E}$: functor on product category. Unraveling definition:

- $F(id_A, id_B) = id_{F(A,B)}$
- $F(f' \circ f,\ g' \circ g) = F(f', g') \circ F(f, g)$

Example: Binary product $(- \times -)\colon \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ (if $\mathcal{C}$ has binary products): on morphisms $f\colon A \to A'$, $g\colon B \to B'$, gives

$$f \times g = \langle f \circ fst,\ g \circ snd \rangle \colon A \times B \to A' \times B'$$

Contravariant and Mixed-Variance Functors

Contravariant functor $F\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$: sends $f : A \to B$ to $F(f)\colon F(B) \to F(A)$; reverses composition: $F(g \circ f) = F(f) \circ F(g)$

Mixed variance $F\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{D} \to \mathcal{E}$: contravariant in first argument, covariant in second

Example: $\mathrm{Hom}(-,-)\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$ is mixed-variance (see next slide)

Duality: contravariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ is the same as covariant functor $\mathcal{C} \to \mathcal{D}^{\mathrm{op}}$; choice of perspective is a matter of convention

Hom-Functor

Hom-functor $\mathrm{Hom}(-,-)\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$: For given $f\colon A' \to A$ and $g\colon B \to B'$, $\mathrm{Hom}(f, g)$ sandwiches input between f and g:

$$\mathrm{Hom}(f,g)\colon \mathrm{Hom}(A,B) \to \mathrm{Hom}(A',B') \qquad \begin{tikzcd} A \arrow[d, "h"'] & & A' \arrow[r, "f"] \arrow[d, dashed] & A \arrow[d, "h"] \\ B & \arrow[r, phantom, "\longmapsto"] & B' & B \arrow[l, "g"] \end{tikzcd}$$

This specializes in two ways:

- $\mathrm{Hom}(A,-)\colon \mathcal{C} \to \mathbf{Set}$ — covariant; acts by postcomposition: $h \mapsto f \circ h$
- $\mathrm{Hom}(-,B)\colon \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ — contravariant; acts by precomposition: $h \mapsto h \circ f$

Contravariant powerset functor sends $X \mapsto \mathcal{P}(X)$ and $f\colon X \to Y$ to $f^{-1}[-]\colon \mathcal{P}(Y) \to \mathcal{P}(X)$

This is isomorphic to $\mathrm{Hom}_{\mathbf{Set}}(X, 2)$ (PROVE IT)

Full and Faithful Functors

Functor $F\colon \mathcal{C} \to \mathcal{D}$ induces maps on hom-sets:

$$F_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A,B) \to \mathrm{Hom}_{\mathcal{D}}(FA,FB), \qquad f \mapsto F(f)$$

- F is faithful if every $F_{A,B}$ is injective
- F is full if every $F_{A,B}$ is surjective
- F is fully faithful if every $F_{A,B}$ is a bijection

Examples:

- Forgetful $U\colon \mathbf{Mon} \to \mathbf{Set}$: faithful (distinct homomorphisms remain distinct) but not full (not every function is a homomorphism)
- Inclusion $(P,\leq) \hookrightarrow \mathbf{Set}$ via poset-as-category: faithful (Hom has at most one element), not full in general

Natural Transformations

Definition: Natural Transformation

For functors $F, G\colon \mathcal{C} \to \mathcal{D}$, natural transformation α assigns to each object X morphism $\alpha_X\colon F(X) \to G(X)$ (component at X) such that for every $f : X \to Y$ naturality square commutes:

$$\begin{tikzcd} F(X) \arrow[r, "F(f)"] \arrow[d, "\alpha_X"'] & F(Y) \arrow[d, "\alpha_Y"] \\ G(X) \arrow[r, "G(f)"'] & G(Y) \end{tikzcd}$$

$$G(f) \circ \alpha_X \;=\; \alpha_Y \circ F(f)$$

- Write $\alpha\colon F \Rightarrow G$
- α is a natural isomorphism if each α_X is an isomorphism; write $F \cong G$

Programming intuition: Naturality $\approx$ parametric polymorphism — a parametrically polymorphic function $\forall A.\, F(A) \to G(A)$ is automatically natural

Examples: Natural Transformations

Natural Transformations between Endofunctors on Set:

- Identity: $id\colon \mathrm{Id} \Rightarrow \mathrm{Id}$ — trivially natural
- Singleton: $sngl\colon \mathrm{Id} \Rightarrow \mathrm{List}$, $sngl_X(x) = [x]$:

$$\begin{tikzcd}
x \in X \arrow[rrr, mapsto, "f"] \arrow[ddd, mapsto, "sngl_X"'] & & & f(x) \arrow[ddd, mapsto, "sngl_Y"] \\
& X \arrow[r, "f"] \arrow[d, "sngl_X"'] & Y \arrow[d, "sngl_Y"] & \\
& \mathrm{List}(X) \arrow[r, "\mathrm{List}(f)"'] & \mathrm{List}(Y) & \\
{[x]} \arrow[rrr, mapsto, "\mathrm{List}(f)"'] & & & {[f(x)]}
\end{tikzcd}$$

- Flatten: $\mu\colon \mathrm{List}\,\mathrm{List} \Rightarrow \mathrm{List}$, $\mu_X([[a,b],[c]]) = [a,b,c]$ — natural (PROVE IT)
- Reverse: $rev\colon \mathrm{List} \Rightarrow \mathrm{List}$, $rev_X([x_1,\ldots,x_n]) = [x_n,\ldots,x_1]$ — natural (PROVE IT)

Non-Example: Sort is Not Natural

Let sort_X sort $\mathsf{List}(X)$ by some total order on X

Counterexample: Let $f : \{a, b\} \to \{a, b\}$ swap a and b where $a < b$:

$$
\begin{tikzcd}
{[b,a]} \arrow[rrr, mapsto, "\mathsf{List}(f)"] \arrow[ddd, mapsto, "\mathit{sort}_X"'] & & & {[a,b]} \arrow[ddd, mapsto, "\mathit{sort}_Y"] \\
& \mathsf{List}(X) \arrow[r, "\mathsf{List}(f)"] \arrow[d, "\mathit{sort}_X"'] & \mathsf{List}(Y) \arrow[d, "\mathit{sort}_Y"] & \\
& \mathsf{List}(X) \arrow[r, "\mathsf{List}(f)"'] & \mathsf{List}(Y) & \\
{[a,b]} \arrow[rrr, mapsto, "\mathsf{List}(f)"'] & & & \color{red}{[b,a] \neq [a,b]}
\end{tikzcd}
$$

Intuitively, components of natural transformations must be agnostic about properties of objects not preserved by morphisms

Functor Category $[\mathcal{C}, \mathcal{D}]$

- **Objects**: functors $\mathcal{C} \to \mathcal{D}$
- **Morphisms**: natural transformations $F \Rightarrow G$
- **Composition**: for $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$, define $(\beta \circ \alpha)_X = \beta_X \circ \alpha_X$

$$\begin{tikzcd}
F(X) \arrow[r, "F(f)"] \arrow[d, "\alpha_X"'] & F(Y) \arrow[d, "\alpha_Y"] \\
G(X) \arrow[r, "G(f)"] \arrow[d, "\beta_X"'] & G(Y) \arrow[d, "\beta_Y"] \\
H(X) \arrow[r, "H(f)"'] & H(Y)
\end{tikzcd}$$

- **Identity**: $(id_X : X \to X)_X$

Natural Isomorphisms

$\alpha\colon F \Rightarrow G$ is natural isomorphism if each component α_X is isomorphism

- Equivalently: α is isomorphism in $[F, G]$ (PROVE IT)
- F and G are naturally isomorphic, written $F \cong G$, if such α exists

Examples (PROVE IT) :

- Swap: $(- \times -) \cong (- \times -)$ via $\mathit{swap}_{X,Y}\colon X \times Y \to Y \times X$, $\mathit{swap}_{X,Y} = \langle \mathit{snd}, \mathit{fst} \rangle$
- Products are associative: $(X \times Y) \times Z \cong X \times (Y \times Z)$ naturally in X, Y, Z
- In **Set**: $\mathrm{Hom}(1, X) \cong X$ naturally in X (elements as arrows)

Practical Remarks

- Often, instead "α is natural transformation $F \Rightarrow G$" one says "family $\alpha_X : F(X) \to G(X)$ is natural in X"
- Same for bi-functors: $\alpha_{X,Y} : F(X,Y) \to G(X,Y)$ is natural in X and Y, etc
- Naturality in (X,Y) $\Leftrightarrow$ naturality in X + naturality in Y (**PROVE IT**) :

$$\begin{tikzcd}
F(X,Y) \arrow[r, "\alpha_{X,Y}"] \arrow[d, "F(f,Y)"'] & G(X,Y) \arrow[d, "G(f,Y)"] \\
F(X',Y) \arrow[r, "\alpha_{X',Y}"'] & G(X',Y)
\end{tikzcd}
\qquad
\begin{tikzcd}
F(X,Y) \arrow[r, "\alpha_{X,Y}"] \arrow[d, "F(X,g)"'] & G(X,Y) \arrow[d, "G(X,g)"] \\
F(X,Y') \arrow[r, "\alpha_{X,Y'}"'] & G(X,Y')
\end{tikzcd}$$

Further Remarks

Composition of natural transformations and functors (whiskering):

- $K\alpha\colon (K\alpha)_X = K(\alpha_X)$ (post-whiskering)
- $\beta H\colon (\beta H)_X = \beta_{H(X)}$ (pre-whiskering)

Despite symmetry: $K\alpha$ is α wrapped in K, while βH — alternatively β_H — only varies component of β. So, more suggestively:

$$(K\alpha)_X = K(\alpha_X) \qquad (\beta_H)_X = \beta_{HX}$$

Often, natural transformations are conflated with their components, e.g:

$$\begin{tikzcd} & K(F(X)) \arrow[dr, "K\alpha"] & \\ H(F(X)) \arrow[ur, "\beta"] \arrow[dr, "H\alpha"'] & & K(G(X)) \\ & H(G(X)) \arrow[ur, "\beta"'] & \end{tikzcd}$$

Natural Transformations

Exponentials

Hom's v.s. Exponentials

- External Hom: $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ lives outside $\mathcal{C}$ — $\mathrm{Hom}\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$
- Internal Hom = Exponential object Y^X — representing same morphisms inside $\mathcal{C}$:

$$(-)^{(-)}\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{C}$$

- **Example in Pos**:
  - $\mathrm{Hom}_{\mathbf{Pos}}(X, Y)$ = monotone maps $X \to Y$ — set, outside **Pos**
  - Y^X in **Pos** = same maps with pointwise order: $f \leq g \iff \forall x.\, f(x) \leq g(x)$ — inside **Pos**

Exponential Object: Definition

Fix object X in a category $\mathcal{C}$ with binary products

- **Object**: Y^X (also written $[X, Y]$ or $X \Rightarrow Y$)
- **Diagram**: evaluation morphism $ev_{X,Y} : Y^X \times X \to Y$
- **Universal property**: for any $f : Z \times X \to Y$, there exists a unique $curry_{X,Y,Z}(f) : Z \to Y^X$ such that:

$$\begin{tikzcd}
Z \times X \arrow[rr, "curry_{X,Y,Z}(f) \times id_X"] \arrow[drr, "f"'] & & Y^X \times X \arrow[d, "ev_{X,Y}"] \\
& & Y
\end{tikzcd}$$

Same caveat as before about "an exponential" and "selected exponential" applies

Cartesian Closed Categories

Category $\mathcal{C}$ is cartesian closed (CCC) if it has:

- terminal object 1
- binary products $X \times Y$ for all X, Y
- exponentials Y^X for all X, Y

CCC are exactly categorical models of typed λ-calculus (with product types)!

Examples:

- Easy: **Set** – suitable for semantics of higher-order without divergence (cf. Paul's course)
- Hard: categories of domains (e.g. Scott domains) for modeling divergence

Axioms of Exponentials

For selected exponentials, one can show (PROVE IT) :

$$(\beta)\quad ev \circ (curry(f) \times id) = f$$

$$(\eta)\quad curry(ev) = id$$

$$(\mathrm{nat})\quad curry(f) \circ g = curry(f \circ (g \times id))$$

Theorem: This axiomatization is sound and complete(PROVE IT)

Exponential Object: Basic Properties

- For $g\colon Z \to Y^X$, define $\mathit{uncurry}(g)$ as

$$Z \times X \xrightarrow{g \times id_X} Y^X \times X \xrightarrow{ev_{X,Y}} Y$$

- Then, $\mathit{uncurry} = \mathit{curry}^{-1}$, witnessing isomorphism (PROVE IT)

$$\mathrm{Hom}(Z \times X, Y) \cong \mathrm{Hom}(Z, Y^X) :$$

- curry, $\mathit{uncurry}$ are natural in all involved parameters, e.g. (nat) is precisely naturality of $\mathit{curry}_{X,Y,Z}$ in Z:

$$\begin{tikzcd}
\mathrm{Hom}(Z \times X,\ Y) \arrow[r, "curry"] \arrow[d, "- \circ (h \times id_X)"'] & \mathrm{Hom}(Z,\ Y^X) \arrow[d, "- \circ h"] \\
\mathrm{Hom}(Z' \times X,\ Y) \arrow[r, "curry"'] & \mathrm{Hom}(Z',\ Y^X)
\end{tikzcd}$$

Examples of Exponentials

- **Set**: Y^X = set of functions $X \to Y$; $\mathit{ev}(g, x) = g(x)$; $\mathit{curry}(f)(z)(x) = f(z, x)$
- **Pos**: Y^X = only monotone functions $X \to Y$ (!) with pointwise order:

$$g \leq h \;\Leftrightarrow\; \forall x.\, g(x) \leq h(x)$$

  curry and ev are defined by same expressions
- Poset categories: recall, $x \to y$ iff $x \leq y$
  - exponential $y^x = (x \Rightarrow y)$ is the Heyting implication: largest z with $z \wedge x \leq y$
  - ev is modus ponens $z \wedge (z \Rightarrow y) \leq y$
- **Non-Example:** in **Mon**, exponentials generally do not exist (PROVE IT)

Natural Transformations

Yoneda Theorem (Lemma)

Polynomial Functors

Polynomial functors are functors of the form:

$$P(X) \;=\; \coprod_{i \in I} X^{n_i} \qquad n_i \in \mathbb{N}$$

where $X^n = \underbrace{X \times \cdots \times X}_{n}$ (powers) and $\coprod_{i \in I}$ are iterated products and coproducts

Example: Polynomial functors model algebraic signatures. For example, monoid $(M, e, \cdot)$ (in **Set**) is represented by morphism (function)

$$f\colon P(M) \to M \qquad \text{where} \qquad P(X) = 1 + X \times X$$

Recall, necessarily $f = [f_1, f_2]$ with $f_1\colon 1 \to M$, $f_2\colon M \times M \to M$. $f_1(\star) = e$, $f_2(x, y) = x \cdot y$

Example: List $X = 1 + X + X^2 + \ldots = \coprod_{n \in \mathbb{N}} X^n$ (needs countable coproducts)

“Polymorphic” List Transformations

Problem: Compute Nat(List, List)

For example,

- $\mathit{rev} \in \mathrm{Nat}(\mathrm{List}, \mathrm{List})$
- $(xs \mapsto [\,]) \in \mathrm{Nat}(\mathrm{List}, \mathrm{List})$
- $(\mathit{case}\; xs\;\; \mathit{of}\;\; [\,] \mapsto [\,]; [y \mid ys] \mapsto ys) \in \mathrm{Nat}(\mathrm{List}, \mathrm{List})$

Natural transformations $P \Rightarrow Q$ are “polymorphic operators” from P to Q

Lemma (PROVE IT) :

$$\mathrm{Nat}(P + R, Q) \cong \mathrm{Nat}(P, Q) \times \mathrm{Nat}(R, Q) \qquad \mathrm{Nat}\left(\coprod_{i \in I} P_i,\; Q\right) \cong \prod_{i \in I} \mathrm{Nat}(P_i,\; Q)$$

Yoneda for Powers

- By previous lemma:

$$\mathrm{Nat}(\mathrm{List},\mathrm{List}) = \mathrm{Nat}\Big(\coprod_{n\in\mathbb{N}} \mathrm{Id}^n, \mathrm{List}\Big) \cong \prod_{n\in\mathbb{N}} \mathrm{Nat}(\mathrm{Id}^n, \mathrm{List})$$

- But how to compute $\mathrm{Nat}(\mathrm{Id}^n, \mathrm{List})$? More generally, how to compute $\mathrm{Nat}(\mathrm{Id}^n, F)$ for $F\colon \mathbf{Set} \to \mathbf{Set}$?
- Take $\alpha\colon\ \mathrm{Id}^n \Rightarrow F$. Then $\alpha_n\colon n^n \to F(n)$, so we can obtain $\alpha_n(id_n\colon n \to n) \in F(n)$
- Conversely: take $a \in F(n)$. Then $\alpha_X\colon X^n \to FX$, sending

$$(f\colon n \to X) \quad \mapsto \quad (F(f)\colon Fn \to FX)(a)$$

  is natural transformation (**PROVE IT**)
- These transitions form isomorhism (**PROVE IT**)

$$\mathrm{Nat}(\mathrm{Id}^n,\ F)\ \cong\ F(n)$$

Natural List Transformers

Combine results of the previous slide with $F = \mathrm{List}$:

$$\mathrm{Nat}(\mathrm{List}, \mathrm{List}) \;\cong\; \prod_{n\in\mathbb{N}} \mathrm{Nat}(\mathrm{Id}^n, \mathrm{List}) \;\cong\; 0^0 \times 1^1 \times 2^2 \times \ldots$$

Action of $t \in \left([\,], [1], [k_1^2, k_2^2], [k_1^3, k_2^3, k_3^3], \ldots\right)$:

$$[x_1, \ldots, x_n] \in \mathrm{List}(X) \quad \mapsto \quad [x_{k_1}, \ldots, x_{k_n}] \in \mathrm{List}(X)$$

Examples (Revisited):

- rev: $t = ([\,], [1], [2, 1], [3, 2, 1], \ldots)$
- $(xs \mapsto [\,])$: $t = ([\,], [\,], [\,], \ldots)$
- $(\mathit{case}\; xs\;\; \mathit{of}\;\; [\,] \mapsto [\,]; [y \mid ys] \mapsto ys)$: $t = ([\,], [\,], [2], [2, 3], \ldots)$

Yoneda Theorem

Theorem: For any locally small $\mathcal{C}$, $F : \mathcal{C} \to \mathbf{Set}$, object $A \in \mathcal{C}$:

$$\mathrm{Nat}(\mathrm{Hom}_{\mathcal{C}}(A, -),\ F) \ \cong\ F(A)$$

naturally in F and A:

$$\begin{tikzcd} \mathrm{Nat}(\mathrm{Hom}(A,-),F) \arrow[r, "\cong"] \arrow[d, "\sigma\circ(-)"'] \arrow[dr, phantom, "\sigma : F \Rightarrow G"] & F(A) \arrow[d, "\sigma_A"] \\ \mathrm{Nat}(\mathrm{Hom}(A,-),G) \arrow[r, "\cong"'] & G(A) \end{tikzcd}$$

$$\begin{tikzcd} \mathrm{Nat}(\mathrm{Hom}(A,-),F) \arrow[r, "\cong"] \arrow[d, "\alpha\mapsto\alpha\circ\mathrm{Hom}(f,-)"'] \arrow[dr, phantom, "f : A \to B"] & F(A) \arrow[d, "F(f)"] \\ \mathrm{Nat}(\mathrm{Hom}(B,-),F) \arrow[r, "\cong"'] & F(B) \end{tikzcd}$$

Under: $\alpha : (\mathrm{Hom}_{\mathcal{C}}(A, -) \Rightarrow F) \ \mapsto\ \alpha_A(id_A) \qquad (f : A \to B \mapsto F(f)(a))_B \ \leftarrow\!\shortmid\ a \in F(A)$

Yoneda Embedding

Yoneda Embedding: $\mathrm{y}\colon \mathcal{C}^{\mathrm{op}} \to [\mathcal{C}, \mathbf{Set}]$ by $\mathrm{y}(A) = \mathrm{Hom}_{\mathcal{C}}(A, -)$

Corollary I: y is faithful: if $\mathrm{y}(f) = \mathrm{y}(g)$ then $f = g$

Proof: Isomorphism

$$\mathrm{Nat}(\mathrm{Hom}(A,-),\ \mathrm{Hom}(B,-)) \ \cong\ \mathrm{Hom}(B,A) \tag{$*$}$$

sends $y(f)$ and $y(g)$ to $\mathrm{Hom}(f,A)(id_A) = f$ and $\mathrm{Hom}(g,A)(id_A) = g$

Corollary II: y is full: every $\alpha\colon \mathrm{Hom}(A,-) \Rightarrow \mathrm{Hom}(B,-)$ equals $\mathrm{y}(f)$ for some $f\colon B \to A$

Proof: Isomorphism $(*)$ sends α to $\alpha_A(id_A)$ and $y(f)$ to f where $f = \alpha_A(id_A)$

Dually (co-Yoneda): take $\mathcal{C} := \mathcal{C}^{\mathrm{op}} \quad \rightsquigarrow \quad y'\colon \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ by $\mathrm{y}'(A) = \mathrm{Hom}_{\mathcal{C}}(-, A)$

Yoneda for Posets

Let $(P, \leq)$ be poset category. Then

- $\mathrm{Hom}(-, a) = {\downarrow}a = \{b \mid b \leq a\} \colon P^{\mathrm{op}} \to \mathbf{Set}$ (principal ideals)
- $\alpha \colon \mathrm{Hom}(-, a) \Rightarrow \mathrm{Hom}(-, b)$ is wittiness of ${\downarrow}a \subseteq {\downarrow}b$

Co-Yoneda Embedding: $y' \colon P \to [P^{\mathrm{op}}, \mathbf{Set}]$, $a \mapsto {\downarrow}a$, factors through downset completion:

$$P \xrightarrow{a \mapsto {\downarrow}a} [P^{\mathrm{op}}, 2] = \{D \subseteq P \mid D \text{ is down-closed}\} \hookrightarrow [P^{\mathrm{op}}, \mathbf{Set}]$$

Faithfulness and Fullness: ${\downarrow}a = {\downarrow}b \;\Rightarrow\; a = b, \quad {\downarrow}a \subseteq {\downarrow}b \iff a \leq b$

Upshot: $[P^{\mathrm{op}}, 2]$ = free completion of P to complete lattice (all joins freely added); For general categories: $[C^{\mathrm{op}}, \mathbf{Set}]$ = free completion of C under all colimits

To Be Continued …